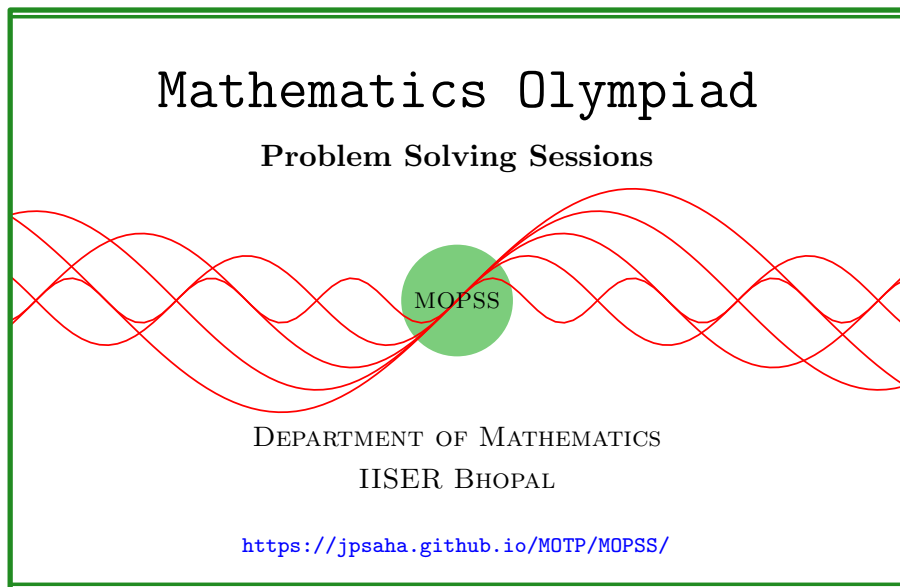


Differentiation and multiple roots

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1 Example (Putnam 1956 B7, IMOSL 1981 Cuba) 2

§1 Differentiation and multiple roots

Lemma 1

Let $P(x)$ be a polynomial with complex coefficients, and α be a complex number. Then α is a root of $P(x)$ having multiplicity at least $r \geq 2$ (i.e., $(x - \alpha)^r$ divides $P(x)$) if and only if it is a root of $P(x), P'(x), \dots, P^{(r)}(x)$, where $P^{(r)}(x)$ denotes the r -fold derivative of $P(x)$.

Solution 1. ■

Example 1.1 (Putnam 1956 B7, IMOSL 1981 Cuba). The polynomials $P(z)$ and $Q(z)$ with complex coefficients have the same set of numbers for their zeroes but possibly different multiplicities. The same is true of the polynomials $P(z) + 1$ and $Q(z) + 1$. **Assume that at least one of $P(z), Q(z)$ is nonconstant.** Prove that $P(z) = Q(z)$.

Walkthrough —

- (a) Assume that $\deg P \geq \deg Q$.
 (b) Denote these two set of roots by S_1, S_2 . Considering multiplicities, show that

$$2 \deg P - |S_1| - |S_2| \leq \deg P' = \deg P - 1,$$

which yields

$$|S_1| + |S_2| > \deg P.$$

- (c) Note that $P - Q$ vanishes at the elements of $S_1 \cup S_2$, which has size larger than the degree of $P - Q$.

Solution 2. On the contrary, let us assume that $P \neq Q$. Without loss of generality, let us assume that $\deg P \geq \deg Q$. Let S_1 (resp. S_2) denote the common set of zeroes of P, Q (resp. $P + 1, Q + 1$). For a polynomial $f(x)$, let us denote its multiset of zeroes by $\mathcal{Z}(f)$.

Note that $\mathcal{Z}(P')$ contains $\mathcal{Z}(P) \setminus S_1$, and $\mathcal{Z}(P')$ also contains $\mathcal{Z}(P + 1) \setminus S_2$. Since $\mathcal{Z}(P)$ and $\mathcal{Z}(P + 1)$ are disjoint, it follows that

$$2 \deg P - |S_1| - |S_2| \leq \deg P' < \deg P - 1,$$

where the final step holds since $\deg P \geq \deg Q$, and one of P, Q is nonconstant. This gives that $|S_1| + |S_2| > \deg P$.

Note that S_1, S_2 are disjoint, and

$$P(z) - Q(z) = (P(z) + 1) - (Q(z) + 1)$$

holds. It follows that $P - Q$ vanishes at $S_1 \cup S_2$, and hence $\deg P \geq |S_1| + |S_2|$. This contradicts the inequality $|S_1| + |S_2| > \deg P$. Consequently, we obtain $P = Q$. ■

References

- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)